

ESP32-C3 Digital Business Card

A rechargeable, NFC-enabled e-paper business card — designed from a blank schematic sheet to a manufacturable PCB

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Abstract

This document records the complete hardware design of a digital business card built around the ESP32-C3: a 100×60 mm, two-layer PCB combining an always-on e-paper display, an NFC tap-to-share interface with a custom-tuned PCB antenna, and USB-C rechargeable Li-Po power. It was designed by hand in KiCad as both a portfolio piece and a cost-competitive alternative to commercial NFC cards. Beyond the final design, this document preserves the *reasoning*: every major component choice traces to a named datasheet section, and the places where the design deliberately deviates from a datasheet's typical application are explained in full, so the trade-offs remain understandable to both an experienced reviewer and my future self.

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1 Introduction

The idea: a business card you never throw away. A phone tapped against the card receives a resume or website link over NFC; an e-paper panel shows a QR code, the date, and phone-set reminders while consuming nothing to hold the image; a small Li-Po recharges over USB-C. Commercial NFC cards sell upward of \$70 with far less functionality, which made this an attractive target for a fully custom board.

Working method. The schematic and layout were built by hand in KiCad, with Claude (Anthropic’s AI assistant) working alongside as senior-engineer reviewer, design-audit partner, and co-author of the project tooling and this document (see the Acknowledgements). Two disciplines were enforced throughout and shaped this document:

- Every design decision traces to a named datasheet or specification section — never to a vendor summary or an unverified claim.
- Everything is verified against primary sources; several plausible-sounding review suggestions were rejected precisely because the primary source said otherwise (Section 7).

Parts are sourced on JLCPCB/LCSC with an LCSC field on every schematic symbol, targeting JLCPCB fabrication with full SMT assembly.

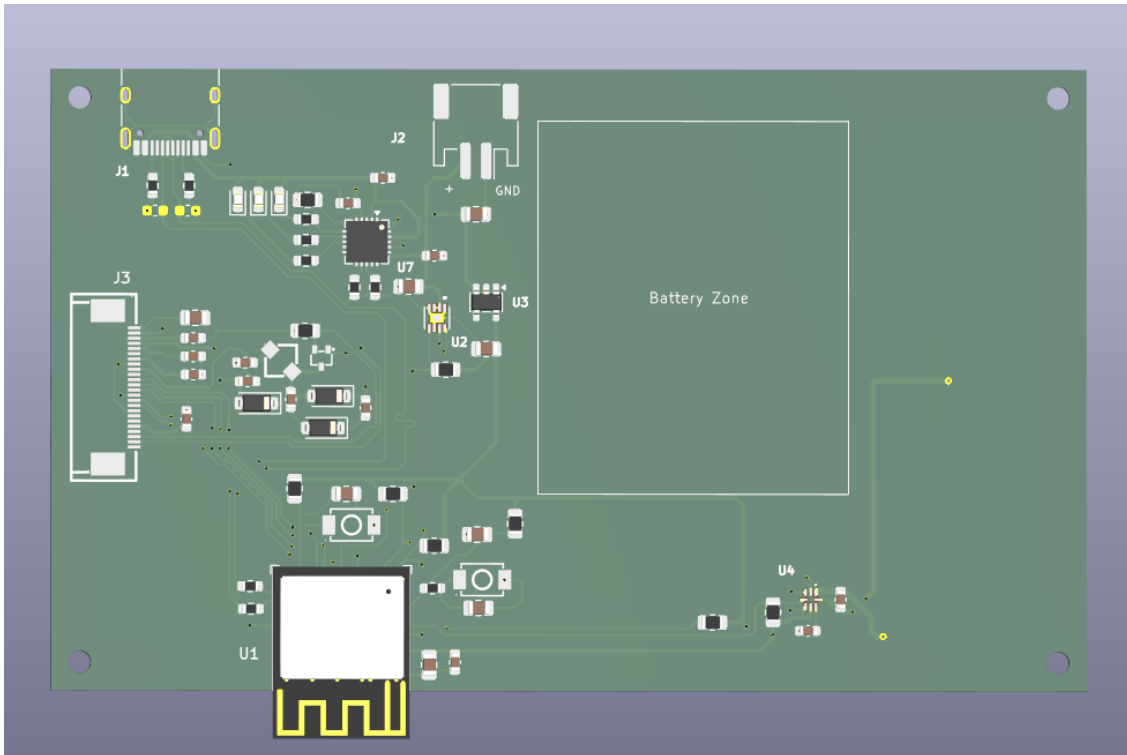


Figure 1: 3D render of the fully populated board (component side), showing the ESP32-C3 module with its antenna overhanging the board edge, the USB-C connector, the e-paper FPC connector (J3), the battery zone, and the charging and power-management cluster.

2 System Overview

2.1 Specifications

2.2 Power architecture

The device spends most of its life in deep sleep with short active bursts (hourly display refresh, NFC taps, BLE sessions) — this duty cycle is what drives the e-paper-centric, low-power archi-

Spec	Value
Board	100 × 60 mm, 2-layer, single-sided assembly
MCU	ESP32-C3-MINI-1-N4 (RISC-V, Wi-Fi + BLE, native USB)
Display	2.66" e-paper, 152×296 px, SSD1680 driver (GDEM0266T71WT)
Connectivity	NFC (NT3H2111 + custom PCB antenna), Wi-Fi, BLE
Charging	USB-C, MCP73871-2CC power-path Li-Po charge management
Regulation	AP2112K-3.3 LDO
Fuel gauge	MAX17048
Battery	1S Li-Po, 200 mA h to 600 mA h (Uxney 503035, 500 mA h selected)
Design tool	KiCad
Manufacturing	JLCPCB, 2-layer, full SMT assembly

Table 1: Headline specifications.

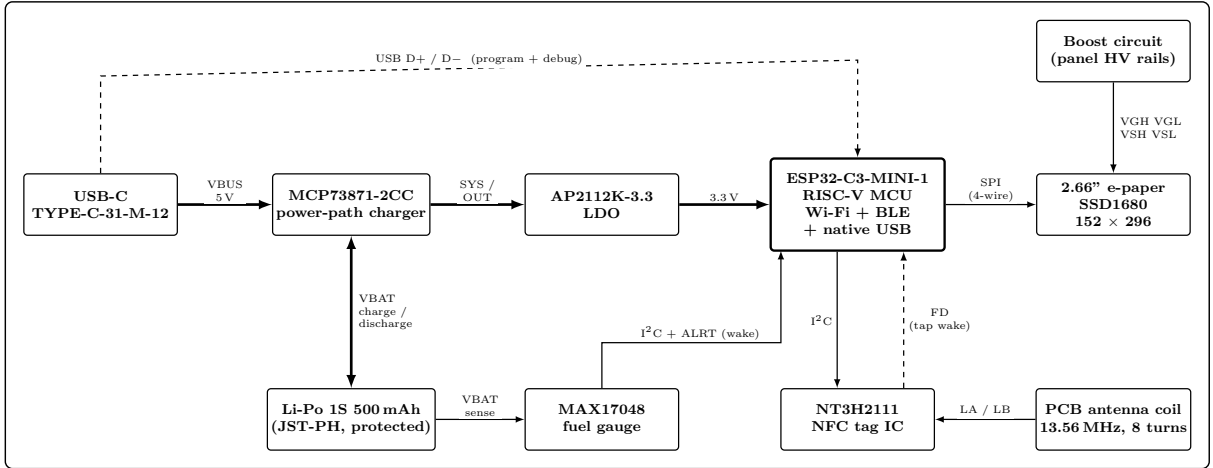


Figure 2: System block diagram. Thick lines carry power; thin lines are digital interfaces; dashed lines are the two deep-sleep wake paths and the USB programming link. The charger’s SYS/OUT rail — not the battery terminal — feeds the LDO, which is what makes the power path seamless.

texture. Mechanically the card is serviceable: a 3D-printed case that opens, with a replaceable JST-PH battery.

3 Background Concepts

The concepts below are used throughout the rest of the document. An experienced reader can skip this section entirely; each later section points back here where a concept first matters.

What is a power-path charger?

A simple charger IC charges the battery, and the system then runs *from the battery* — meaning a dead battery makes the device unusable even when plugged in, and system current confuses the charge-termination logic. A *power-path* charger like the MCP73871 separates the two: the system rail (SYS/OUT) is powered directly from USB when present and switches seamlessly to the battery otherwise, while the battery charges independently. This is why the LDO’s input is the charger’s OUT pin, *not* the battery terminal.

What is a strapping pin?

Some GPIOs are read *at the moment of reset* to configure how the chip boots (normal flash boot vs. download mode, etc.). On the ESP32-C3 these are GPIO2, GPIO8 and GPIO9. If external circuitry pulls one of these the wrong way at power-up, the chip silently boots into the wrong mode. Every strapping pin in this design carries a pull-up that both satisfies boot requirements and serves the attached peripheral (e.g. GPIO2’s pull-up doubles as the e-paper reset idle-high).

What is the 5.1 k CC pull-downs on USB-C?

A USB-C *source* decides how much current to offer by measuring the CC pins. A device that wants power (a “sink”) must present 5.1 k Ω on both CC lines — without them, a USB-C-to-USB-C cable delivers no VBUS at all and the board would mysteriously only charge from A-to-C cables.

What is a DNP (Do Not Populate) part — and why the USB ESD caps are DNP?

DNP means the footprint exists on the board but no part is fitted at assembly: the position is a free option for later. Here, C5/C6 are small capacitor positions on the USB data lines, reserved for ESD/filtering. They are left unpopulated because extra capacitance on USB data lines degrades signal edges, and the ESP32-C3’s internal protection is adequate for a personal device — but the pads cost nothing, so a future revision (e.g. a sale version adding proper TVS diodes) can populate them without a board respin. DNP positions must be excluded when exporting the BOM, or the assembler will try to source a part literally valued “C”.

What is a 0 Ω resistor?

A resistor with (nominally) zero resistance — electrically just a wire, so why fit one? Because it is a *removable, replaceable* wire: it provides a break point to isolate a circuit section during debugging (desolder one part instead of cutting copper), and a substitution point if a small series resistance or ferrite bead turns out to be needed later. In this design, 0 Ω resistors sit in series with the USB D+/D– lines, placed symmetrically on the pair so both data lines see identical parasitics.

What is ceramic capacitor dielectrics (C0G vs. X5R/X7R) and DC bias?

Ceramic capacitors trade size for stability. X5R/X7R dielectrics lose a large fraction of their capacitance as DC voltage is applied across them (“DC bias”) and drift with temperature — fine for decoupling, fatal for a resonant circuit, where drifting capacitance *detunes an antenna* away from its frequency. C0G/NP0 dielectric is nearly ideal (no bias effect, tiny tempco), which is why the NFC tuning capacitor is specified as C0G despite costing more per microfarad. The same physics appears at the e-paper VCOM capacitor, where an X7R part is chosen with enough voltage headroom that its *biased* capacitance still meets the requirement (Section 5.6).

4 Component Selection

Function	Part	LCSC	Package
MCU	ESP32-C3-MINI-1-N4	C2838502	module
Charger	MCP73871-2CC	C5121473	QFN-20 4×4 + EP
Fuel gauge	MAX17048G+T10	C2682616	TDFN-8 2×2 + EP
LDO	AP2112K-3.3	C51118	SOT-23-5
NFC	NT3H2111W0FHKH	C710403	XQFN-8
USB-C	HRO TYPE-C-31-M-12	C165948	16-pin receptacle
FPC connector	AFC07-S24ECA-00 (top contact)	C262643	24P, 0.5 mm
Battery conn.	JST S2B-PH-SM4-TB	C295747	PH 2.0 mm, horiz. SMT
Boost NMOS	SI1304BDL	C7419947	SOT-323
Boost inductor	CDRH2D18, 47 μ H	C2454210	SMD
Schottky ×3	MBR0530	C77336	SOD-123
Buttons ×2	TS-1088-AR02016	C720477	4×3 mm tactile

Table 2: Principal components. The full BOM including all passives is in Appendix C.

Why these parts, briefly:

- **ESP32-C3-MINI-1**: integrated PCB antenna (RF pre-certified, no RF layout risk), native USB for programming and debug with no UART bridge chip, RISC-V core, deep-sleep in the microamp range.
- **MCP73871-2CC**: true power-path management (see explainer above). The *-2CC* suffix specifically encodes 4.20 V regulation, a 6-hour safety timer, and a 3.1 V low-battery threshold — the suffix table on the datasheet’s Product Identification page is the only place these differ between variants.
- **MAX17048**: battery percentage without a sense resistor (ModelGauge), 3 μ A operating current, I²C, and an ALRT pin used as a deep-sleep wake source.
- **NT3H2111**: NFC Forum Type 2 tag with an I²C host interface, energy harvesting, and a field-detect output that wakes the ESP32 from deep sleep on a phone tap.
- **SSD1680 panel (GDEM0266T71WT)**: holds its image at zero power; only refreshes cost energy.

5 Detailed Design

The complete schematic is shown in Figure 3; the subsections below walk through each block and the reasoning behind it.

5.1 MCU: pin map and strapping

The module symbol/footprint come from Espressif’s official KiCad library (<https://github.com/espressif/kicad-libraries>); power, enable and reset networks follow the ESP32-C3-MINI-1 datasheet’s *Peripheral Schematics* section (Espressif recommends $R = 10\text{ k}\Omega$ and $C = 1\text{ }\mu\text{F}$ on EN). Strapping pins — and why they shape this pin map — are explained in Section 3.

Design decision: wake pins must be RTC-capable

Only GPIO0–GPIO5 can wake the ESP32-C3 from deep sleep. Both wake sources were deliberately placed in that range: GPIO3 (NFC field detect — a phone tap wakes the card) and GPIO4 (fuel-gauge alert — a low battery wakes the card to warn the user).

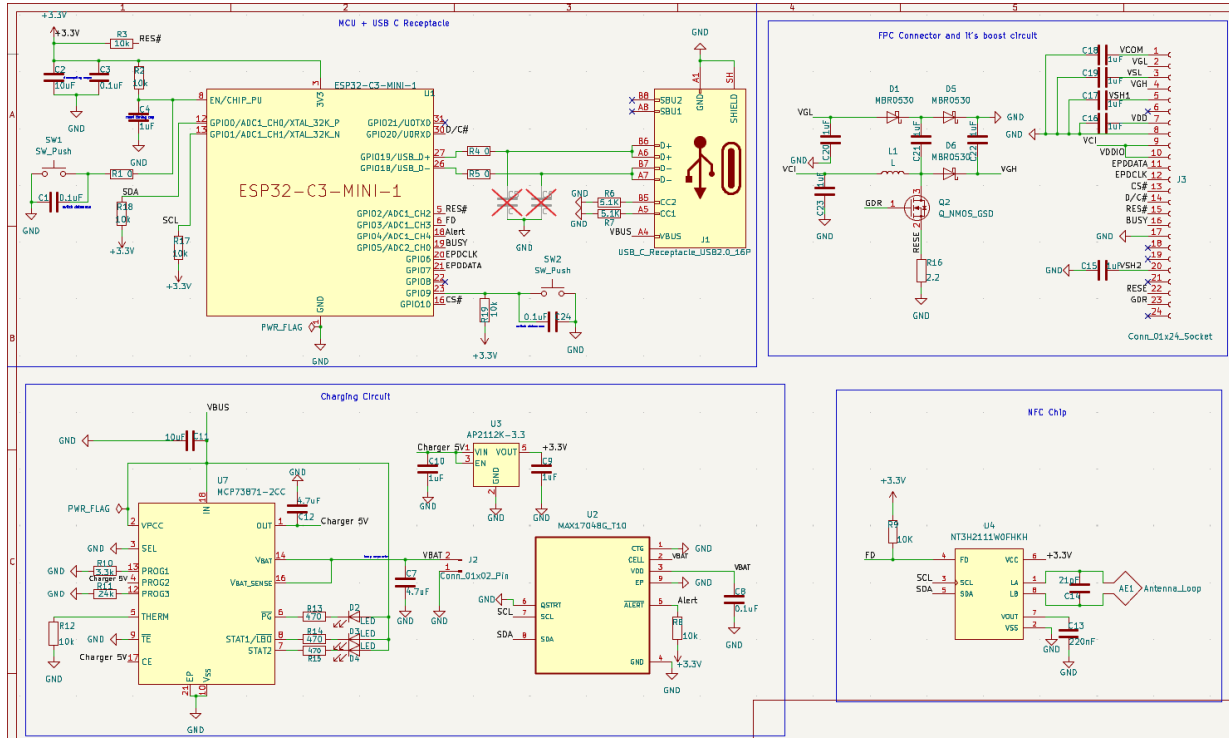


Figure 3: Complete schematic, drawn as four functional blocks: the MCU with its USB-C receptacle (top left), the e-paper FPC connector with its boost circuit (top right), the charging circuit with the LDO and fuel gauge (bottom left), and the NFC front end with its antenna (bottom right).

GPIO	Net	Notes
0	I2C_SDA	shared bus (NFC + fuel gauge)
1	I2C_SCL	shared bus
2	EPD_RST	strapping pin → 10k pull-up, idles high
3	NFC_FD	field detect; deep-sleep wake on tap (RTC-capable)
4	FUELGauge_ALERT	MAX17048 ALERT; deep-sleep wake (RTC-capable)
5	EPD_BUSY	input
6	EPDCLK	e-paper SPI clock
7	EPDDATA	e-paper SPI data
8	STATUS_LED	strapping → must be high at boot, 10k pull-up
9	BOOT button	strapping, 10k pull-up + debounce cap, button to GND
10	EPD_CS	e-paper chip select
18/19	USB D- / D+	0Ω series to USB-C
20	EPD_DC	data/command
21	UART0_TX	optional debug
EN	RESET	10k pull-up + cap + RESET button

Table 3: Boot-safe pin map.

Design decision: renaming the e-paper SPI nets

The panel’s datasheet labels its SPI pins “SCL/SDA” — the same names as the I²C bus, which invites miswiring. The nets were renamed EPDCLK/EPDDATA so the two buses cannot be confused anywhere in the schematic.

5.2 USB-C

HRO TYPE-C-31-M-12 receptacle (USB 2.0, 16-pin). D± route through 0Ω series resistors to GPIO19/18 (the resistors are debug/isolation points placed symmetrically on the pair); CC1/CC2 each carry a 5.1 kΩ pull-down to ground; data-line ESD capacitor positions exist but are DNP. Why the CC pull-downs are essential, why the ESD positions are deliberately unpopulated, and what the 0Ω resistors are for are all covered in Section 3.

The footprint was verified dimension-by-dimension against the connector’s mechanical drawing: mounting-hole spacing 5.78 mm, X-axis pitch 8.9 mm exact, Y placement within the ±0.05 mm drawing tolerance (PCB 7.3 mm vs drawing 7.35 mm), and all pads (4×0.6 mm, 8×0.3 mm) matching in size, naming and numbering.

5.3 Battery charger: MCP73871-2CC

Every configuration strap traces to a datasheet section:

Pin	Strap	Effect (datasheet ref)
SEL	GND	USB mode (§3.4, §5.2.2)
PROG2	OUT	500 mA input limit (§3.9)
PROG1	3.3 kΩ to VSS	≈303 mA charge current (Eq. 5-1)
PROG3	24 kΩ to VSS	≈42 mA termination (Eq. 4-2)
THERM	fixed 10 kΩ to VSS	see below (§5.1.4)
CE	OUT	charge enabled, never floating (§3.16)
TE	GND	6-hour safety timer on
VPCC	IN	disabled; input current limit already caps at 500 mA (§3.3)

Table 4: MCP73871 strap configuration.

Design decision: no thermistor — a fixed 10 k on THERM

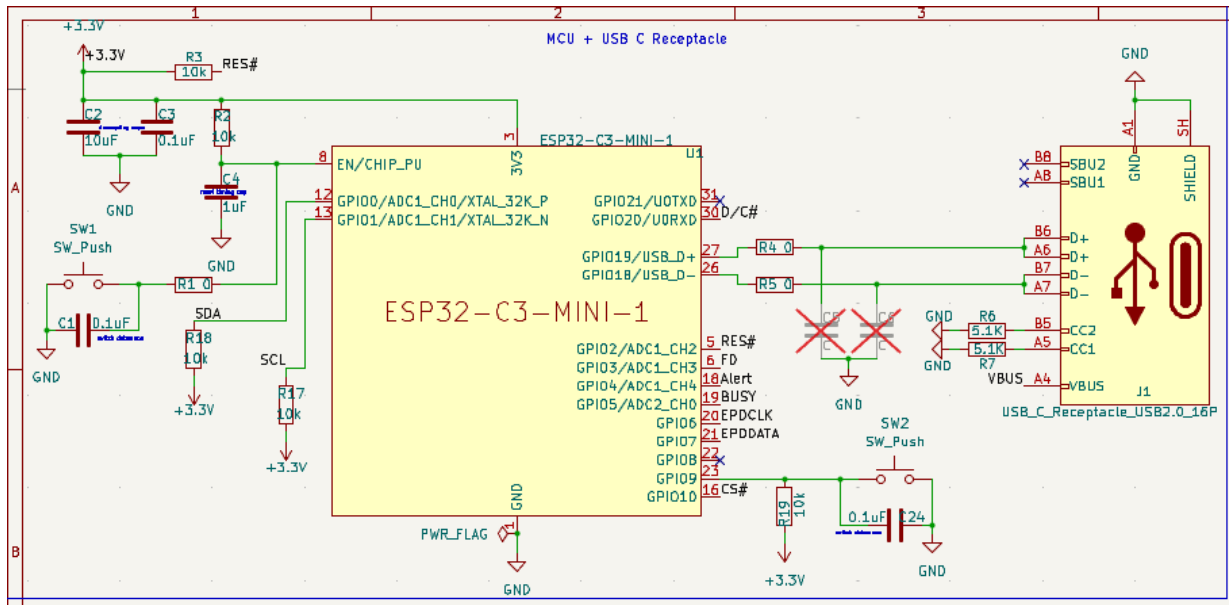
The MCP73871 expects the battery pack to contain an NTC thermistor (a resistor whose value changes with temperature) wired to its THERM pin. The charger pushes a fixed 50 μA out of that pin and measures the voltage that appears; by Ohm’s law ($V = I \times R$) the voltage reveals the thermistor’s resistance, and therefore the battery’s temperature. A reading above 1.24 V means “resistance high → battery too cold”; below 0.25 V means “resistance low → battery too hot” — and the charger pauses charging either way.

This design’s battery is a small protected Li-Po with no thermistor wire available, so THERM instead gets a **fixed 10 kΩ to VSS**: $50 \mu\text{A} \times 10 \text{ k}\Omega = 0.5 \text{ V}$, permanently in the middle of the safe window. The charger is deliberately made to always believe the battery is at a comfortable temperature, and temperature safety is knowingly delegated to the protection circuit built into the battery pack itself.

Two traps live here, and both nearly happened:

- **Grounding THERM** reads 0 V — below the hot threshold, so the charger suspends charging *permanently*. The intuitive “tie the unused pin low” instinct produces a board that never charges.
- **The wrong resistor value does the same.** A BOM error (caught in a pre-order audit) had assigned the 3.3 kΩ part here: $50 \mu\text{A} \times 3.3 \text{ k}\Omega = 0.165 \text{ V}$ — also read as “too hot,” also a card that never charges. The margin analysis above is the whole point of the 10 kΩ value.

Status LEDs (STAT1/STAT2/PG) each drive an LED + 470Ω pulled to VBUS — PG must be referenced to the input rail per §5.2.4, and pulling to VBUS means the LEDs only light when



9 Peripheral Schematics

This is the typical application circuit of the module connected with peripheral components (for example, power supply, antenna, reset button, JTAG interface, and UART interface).

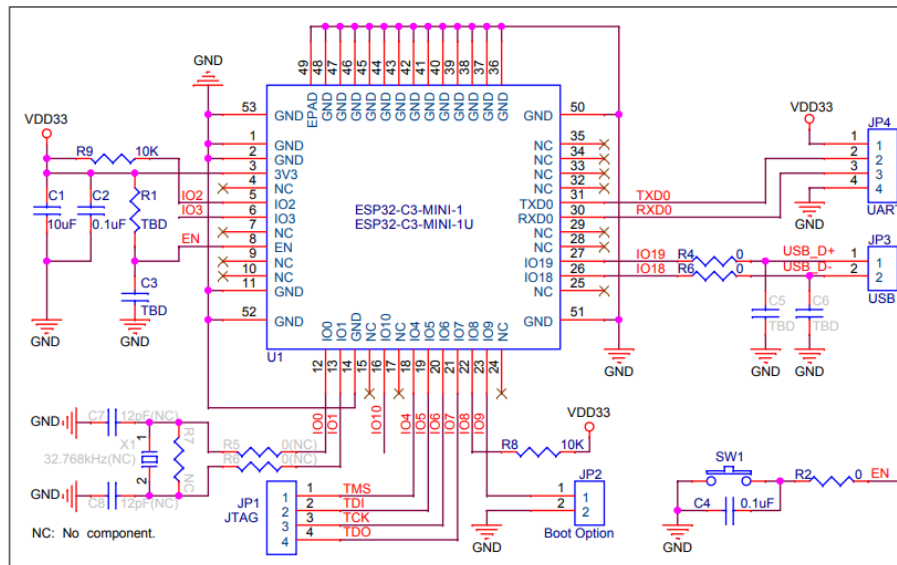


Figure 9-1. Peripheral Schematics

Figure 4: **Top:** the MCU and USB-C block as implemented — EN reset network (R2/C4 with SW1), BOOT strapping (R19 with SW2 and its debounce cap), decoupling on 3V3, and D± through their 0Ω series resistors to GPIO19/18. **Bottom:** the corresponding reference from the ESP32-C3-MINI-1 datasheet §9 *Peripheral Schematics* (Espressif), which recommends 10 kΩ and 1 μF on EN — the values used here. Comparing the two is how the reset, strapping and USB connections were verified rather than assumed.

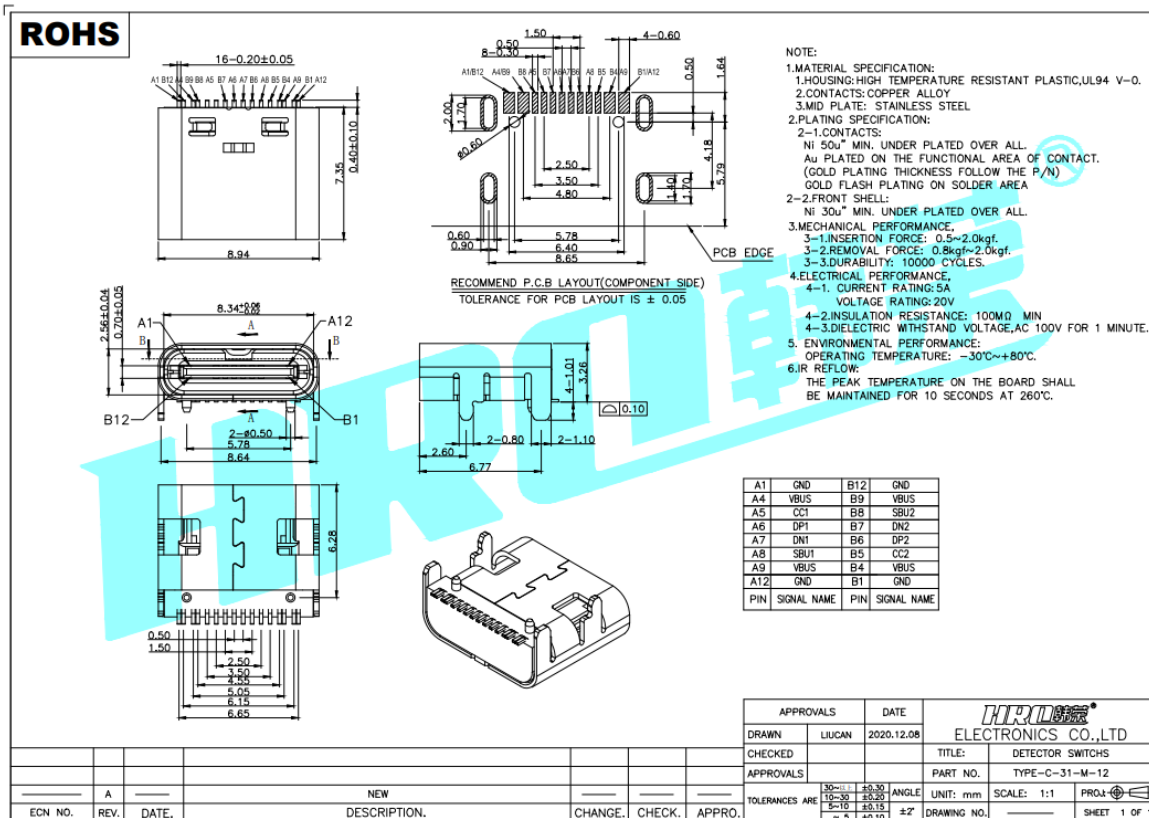
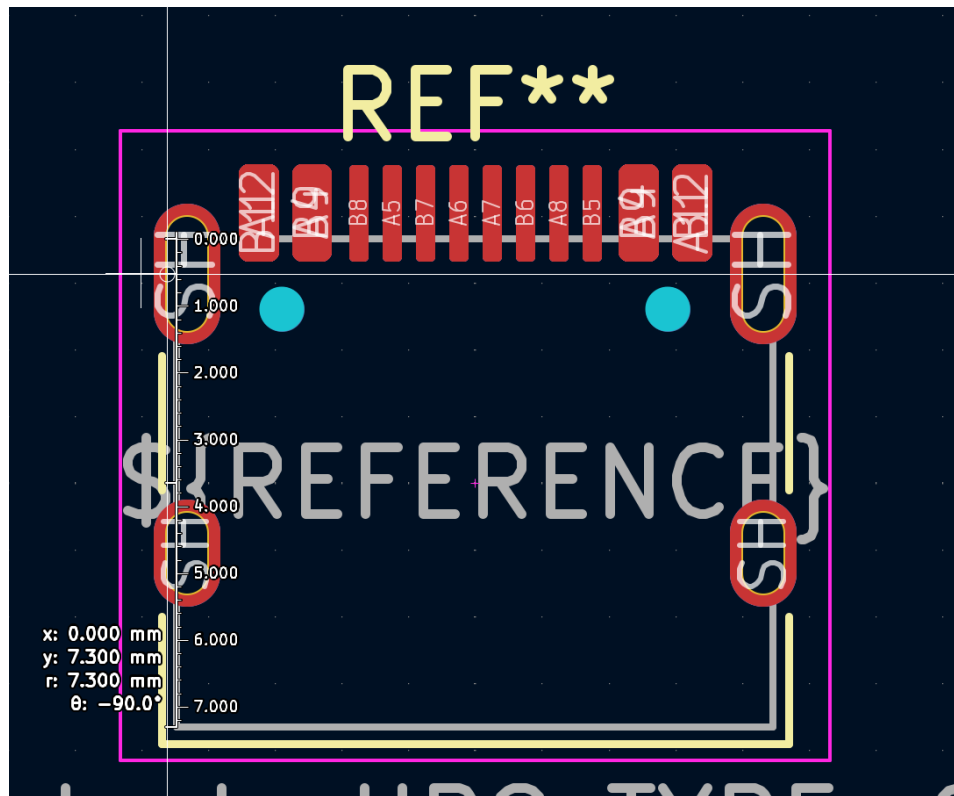


Figure 5: **Top:** the KiCad footprint measured with the built-in measure tool — the readout shows 7.300 mm between the mounting-hole rows. **Bottom:** the corresponding TYPE-C-31-M-12 mechanical drawing (HRO Electronics), whose *Recommended PCB Layout* gives 7.35 mm with a ± 0.05 mm layout tolerance, along with the 5.78 mm and 8.94 mm references and the 4×0.6 mm / 8×0.3 mm pad sizes. Each dimension, pad size and pin name was checked against the drawing before the connector was placed — a library footprint being “the right part name” is not the same as it matching the part you ordered. 10

USB is present, saving battery. The exposed pad ties to GND with thermal vias (the charger dissipates roughly 0.4 W worst-case).

5.4 Regulator and fuel gauge

AP2112K-3.3 (SOT-23-5): input from the charger’s SYS/OUT rail — *not* VBAT, which is the entire point of the power-path architecture — EN tied to VIN, 1 μ F in and out.

MAX17048 (TDFN-8): VDD *and* CELL both connect to VBAT, because VDD is simultaneously the supply and the sense input — powering it from 3.3 V would make it measure the regulator instead of the battery. QSTRT/CTG and the exposed pad go to GND; ALRT goes to GPIO4 as a wake source.

5.5 NFC front end and antenna

The NT3H2111 (XQFN-8, no exposed pad) connects to the shared I²C bus; FD goes to GPIO3 with a pull-up; a 220 nF capacitor sits on VOUT. The antenna connects across LA/LB with the tuning capacitor in *parallel*, forming the resonant tank of NXP AN11276’s equivalent-circuit model.

5.5.1 Antenna design

The antenna is not a component — it is a rectangular planar spiral etched directly into the front copper. The design constants:

- Resonance target: 13.56 MHz (NFC standard)
- Chip internal capacitance: $C_i = 50$ pF (NT3H2111 datasheet §13.1)
- Available area: 15 $\times$ 50 mm strip on the card’s right edge

From the resonance equation

$$f = \frac{1}{2\pi\sqrt{LC}} \Rightarrow L_{\text{ideal}} = \frac{1}{(2\pi f)^2 C_i} \approx 2.755 \mu\text{H}$$

if the chip’s capacitance alone were to resonate the coil. The final geometry (iterated in ST’s eDesign coil tool and cross-checked with an independent Greenhouse-model calculation):

Outer footprint	$\sim 15 \times 50$ mm
Turns	8
Trace width / spacing	0.6 mm / 0.3 mm
Resulting inductance	$\approx 1.93 \mu\text{H}$
Total capacitance for resonance	≈ 71 pF
External tuning cap C14	22 pF C0G/NP0 (LCSC C1653)

Table 5: Final antenna geometry.

Design decision: 8 turns at 0.3 mm spacing, not 6 at 0.6 mm

An earlier 6-turn variant ($\approx 1.15 \mu\text{H}$) would have needed ~ 70 pF of external capacitance. Tightening the spacing to 0.3 mm fits 8 turns in the same strip, raising inductance so only ~ 21 pF external is needed. More of the resonance comes from the coil itself: higher Q, better read range, and less sensitivity to the tuning cap’s tolerance.

The design intentionally sits slightly high in frequency to absorb real-world detuning from nearby metal and the ground pour, and C14 remains an accessible tuning position for bench adjustment (VNA or simple read-range testing) after assembly — C0G dielectric is mandatory in this position (Section 3). The antenna strip is kept clear of copper on *both* faces, and the battery is deliberately placed away from it — a large conductive object behind the coil would shield and

detune it. Because 8 turns leave only a small inner aperture, the inner terminal exits through a via and a single thin bridge on the back copper (a thin bridge is acceptable inside the keep-out; a pour is not).

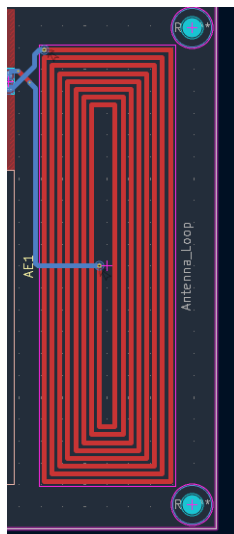


Figure 6: The NFC coil as etched copper in the right-hand strip: 8 rectangular turns at 0.6 mm width and 0.3 mm spacing, with the inner terminal exiting through a via. The surrounding area is kept clear of copper on both layers so the coil couples cleanly to a phone.

5.6 E-paper interface and boost converter

The SSD1680 controller drives the panel over 4-wire SPI (BS1 strapped to GND selects this mode, datasheet §6.3.1) and generates its own high-voltage gate/source/VCOM rails from an external boost circuit, wired one-to-one from the panel vendor’s reference application circuit (GDEM0266T71WT §9): NMOS SI1304BDL, 47 μH CDRH2D18 inductor, three MBR0530 Schottkys, 2.2 Ω sense resistor, 1 μF rail capacitors, and the VCOM capacitor.

Four verification stories from this block are worth preserving:

Design decision: the MOSFET pinout that ERC cannot catch

The generic KiCad NMOS symbol first used had pin order Drain–Gate–Source; the SI1304BDL is Gate–Source–Drain. Every pin was mismatched — a dead boost converter with *zero* ERC warnings, because electrically the netlist was “valid.” Caught only by checking the symbol pin numbers against the datasheet package drawing; fixed by switching to the Q_NMOS_GSD symbol variant and re-verifying all three connections. Symbol-to-package pin mapping is a class of error only datasheet review catches.

Design decision: keeping the reference 47 μH inductor (review challenge rejected)

A review pass claimed the SSD1680 boost “requires 10 μH ” and that the reference part’s 200 mA saturation rating would be exceeded. The panel’s own datasheet settled it: §9 explicitly lists CDRH2D18 47 μH as the reference part, and §6.2 gives actual panel current as 4.5 mA typical operating / 40 mA typical peak — 3–5 $\times$ below saturation even in the strictest reading. The challenge was specific and plausible, but generic; the primary source won. (Documented fallback: a higher- I_{sat} 47 μH part if rails measure low on the bench.)

Design decision: $R_{\text{ds(on)}}$ at the gate drive that actually exists

The SI1304BDL’s headline 450 m Ω is specified at $V_{\text{GS}} = 4.5 \text{ V}$ — a voltage this circuit never reaches. The applicable figure is 0.65 Ω max at $V_{\text{GS}} = 2.5 \text{ V}$, still comfortably under the reference’s $\leq 2.1 \Omega$ requirement. Always read the test conditions column.

Design decision: the VCOM capacitor and DC bias

The reference calls for an effective capacitance $>0.25\mu\text{F}$ at 18 V DC bias in the VCOM position, X7R dielectric required. Verified via Samsung’s MLCC DC-bias data: the chosen $1\mu\text{F}$ 50 V X7R part measures $0.89\mu\text{F}$ at 18.4 V — roughly $3.5\times$ the requirement. The generous voltage headroom (18 V on a 50 V part) is what preserves the capacitance; an X5R part of the same nominal value would not have qualified.

5.6.1 FPC connector: which side are the contacts on?

The panel’s 14 mm tail wraps around the board edge to a back-mounted connector, with the board inside the fold. The panel’s gold fingers are on the screen side — the outer face of the wrap — which stays outer all the way around, so at the connector they face *away* from the board: a **top-contact** connector (AFC07-S24ECA-00) is required. The bottom-contact sibling (S24FCA) differs by one letter and would make the panel physically unconnectable — a one-letter difference worth re-checking every time the BOM is regenerated.

Pin order also reverses through the wrap: connector pad P receives panel pin $(25 - P)$. The layout was verified by matching *signals* (each panel signal landing on the pad carrying that net) rather than trusting pin numbers. BS lands on GND (4-wire SPI select), and VDDIO merges into VCI per the SSD1680 datasheet §5 Table 5-1 — the one intentional ERC informational note in the design.

6 PCB Layout

6.1 Floorplan and stack-up

Two layers, 100×60 mm. The X-axis budget sums exactly: 5 mm margin + 72 mm panel zone + 3 mm divider + 15 mm antenna strip + 5 mm margin = 100 mm.

- **Front:** e-paper panel (held a few millimetres off the board by the case) and the NFC coil in the right-hand strip.
- **Back:** *all* SMD components — keeping assembly single-sided, the cheapest option. The antenna is etched copper (free), and the two vias linking it to the NFC chip do not trigger double-sided assembly charges.
- **Battery:** on the back, behind the panel region — deliberately *not* behind the antenna.

Staying 2-layer was a considered decision: every interface is low-speed (e-paper SPI, I²C, full-speed USB), so a 4-layer stack is not warranted, and only a routability wall would have changed that.

6.2 Grounding on a 2-layer board

There is no hidden plane on a 2-layer board. The ground strategy: components and their pours share the back copper (GND pads connect directly into the fill, no vias needed), while the front carries a large ground pour in the free area under the panel — stopping 3 mm short of the antenna strip. Vias to ground exist only where needed: front-copper returns, stitching, and the thermal arrays under the charger and fuel-gauge exposed pads.

6.3 Module antenna placement

The ESP32-C3-MINI-1 sits at the top edge with its PCB antenna *overhanging* the board by design: Espressif’s hardware design guidelines state the module antenna should be outside the base board with the feed point close to the edge — the preferred configuration for radiation performance, not a compromise. The module body remains fully seated (all 61 pads ≥ 1 mm inboard); only the antenna tail extends past, protected by a keep-out zone on both copper layers.

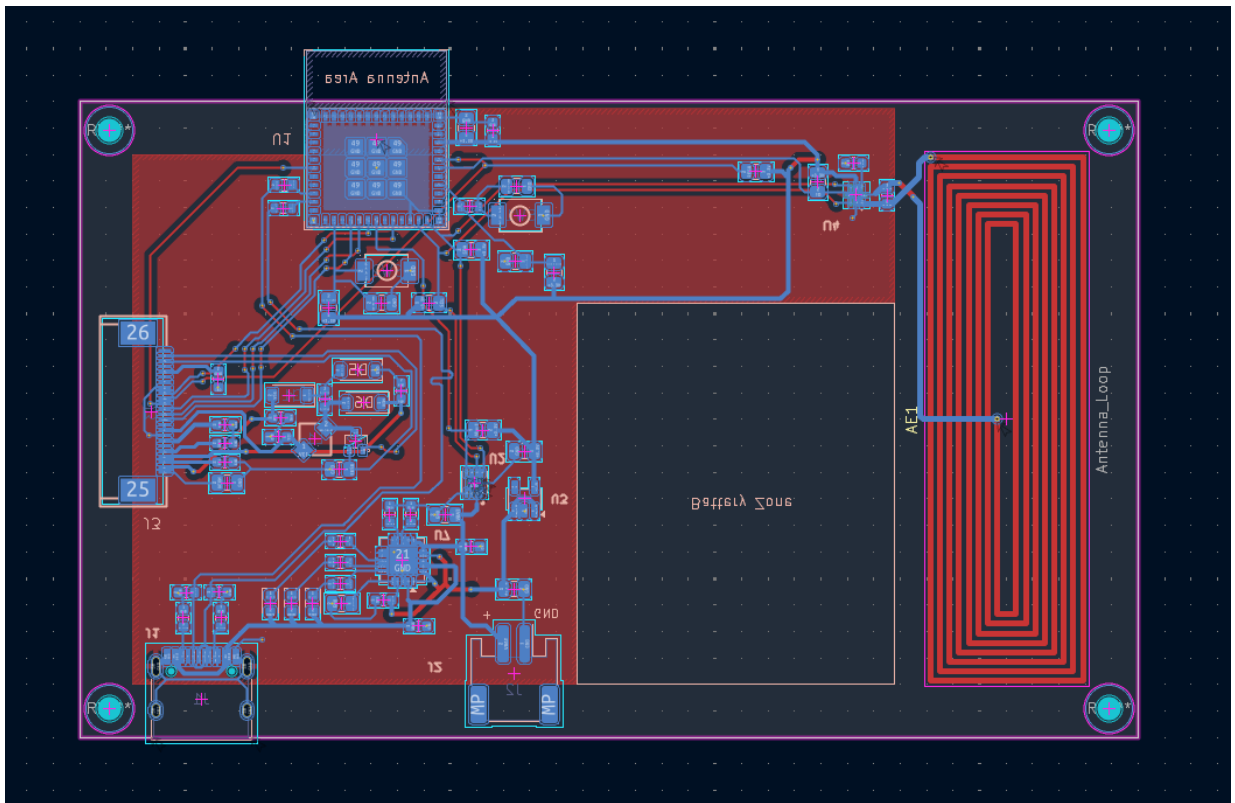


Figure 7: PCB layout showing the functional zone split: components and routing on the back copper (left two-thirds), the reserved battery zone, and the NFC antenna coil in its own strip on the right, with the ground pour kept clear of it on both layers. The ESP32-C3 module sits at the board edge with its antenna area overhanging.

6.4 Net classes and routing discipline

Class	Clearance	Track	Via/drill	Nets
Default	0.2	0.25	0.6/0.3	signals
Power	0.2	0.4	0.6/0.3	3V3, VBUS, VBAT, VCI...
HV	0.5	0.4	0.6/0.3	VGH, VGL, VSH1/2, VSL
USB	0.2	0.25	0.6/0.3	D± pair (0.25/0.2 gap)

Table 6: Net classes (mm). All within JLCPCB standard capability — no fine-feature surcharge.

Routing order: power trunks → USB differential pair → boost switching loop (tight and fat) → NFC feed → HV rails → SPI → I²C → remainder → pours and stitching → DRC. The D± pair measured 0.24 mm length mismatch worst-case — negligible at full-speed USB. Power trunks stay at 0.4 mm, necking only at fine-pitch pad entries.

7 Verification and Design Review

7.1 ERC and DRC

ERC closes clean apart from one documented informational note (VDDIO deliberately tied to VCI per SSD1680 §5). DRC progressed 38 → 12 → 0 violations across review passes, with 12 *documented* exclusions — each with a written justification, because an undocumented exclusion is indistinguishable from a suppressed defect. Representative examples: the MAX17048’s own 0.5 mm-pitch footprint violates the 0.2 mm clearance rule between its own pads (a property of the manufacturer’s land pattern, not the routing); the antenna coil “shorts” against its own feed pads because the spiral is graphic copper with no net; the module silk crosses the board edge because the antenna overhang is intentional.

7.2 Lessons worth keeping

- A net *label* on an unrouted pad looks identical to a connected one in a netlist check — trace actual copper for critical nets (found: two grounded FPC pads with no physical connection, including the BS1 mode strap without which the panel never communicates).
- Plausible, specific-sounding review claims still lose to primary sources (the 10 µH inductor challenge).
- Read electrical characteristics at *your* operating point, not the headline row (R_{ds(on)}, diode V_f, MLCC DC bias).
- Symbol pin mapping vs. package pinout is invisible to ERC — verify against the package drawing (the GSD MOSFET).
- Part-number metadata deserves the same rigor as copper: a final ref-by-ref audit of every LCSC field — value, package *and* variant against the placed footprint — caught substitutions (a wrong THERM resistor, a bottom-contact connector variant) that a copper-perfect design cannot reveal.

8 Manufacturing

Fabrication targets JLCPCB, 2-layer, with full SMT assembly (the 0.5 mm-pitch FPC connector should not be hand-soldered). Gerbers follow JLCPCB’s KiCad guide: Protel extensions, X2 attributes, netlist included, 4.6 mm format; drill files in Excellon. The BOM carries an LCSC **Part #** column and excludes DNP positions (the two ESD capacitor placeholders) and the etched antenna; the CPL (component placement list) gives each part’s centroid, side — all bottom — and rotation.

Pre-order checklist:

- Verify polarized-part orientation (Q2, U7, U2, U4, diodes, LEDs) in JLCPCB’s assembly preview — KiCad and JLCPCB rotation conventions disagree per-part, and the preview is where it shows.
- Nylon (not steel) M2 screws on the two mounting holes nearest the NFC coil, so hardware does not shift the antenna tuning.
- On battery arrival: verify lead polarity against J2 with a multimeter before first connection — JST-PH plug polarity is not universal, and reversal is destructive.

9 Firmware Notes and Future Work

- **Display orientation:** if the mounted panel renders upside down, it is a firmware fix (SSD1680 scan-direction / RAM entry-mode registers 0x01/0x11), not hardware.
- **Temperature:** use the SSD1680’s internal sensor (command 0x18, data 0x80); note it reads die temperature, so displayed ambient will read slightly high — a dedicated sensor is a rev-2 candidate.
- **Flashing:** native USB handles download mode automatically; BOOT+RESET is the guaranteed manual recovery path.
- **NFC tuning:** if read range is marginal, adjust C14 after measuring actual resonance.
- **Rev 2 candidates:** ESD/TVS protection on USB data for the sale version, ambient temperature sensor, expanded thermal via grid under the charger.

Acknowledgements

This project was a collaboration between a human designer and an AI. Claude (Anthropic) acted as the senior-engineer reviewer this project would not otherwise have had: verifying every major decision against the primary datasheets, challenging design assumptions (and gracefully losing the arguments it deserved to lose), auditing the BOM before ordering — where it caught, among other things, the THERM resistor substitution that would have produced a card that never charges — writing the parametric generator that produced the NFC antenna footprint (`tools/make_nfc_antenna.py` → `NFC_Antenna_50x15_8T.kicad_mod`) in place of an error-prone hand-drawn spiral, and drafting the document you are reading. Without that guidance this project may well not have reached a manufacturable state. The design choices — and any mistakes that remain — are mine; several of the catches were not.

A Toolchain and Library Setup

Steps to reproduce the design environment from a clean machine.

A.1 KiCad and the Espressif library

1. Install KiCad (<https://www.kicad.org>). This project was built across KiCad 9/10.
2. Download or clone Espressif’s official KiCad library: <https://github.com/espressif/kicad-libraries>.
3. In KiCad: *Preferences* → *Manage Symbol Libraries* → *Add*, pointing at `Espressif.kicad.sym`; then *Preferences* → *Manage Footprint Libraries* → *Add*, pointing at `Espressif.pretty`. Add the library’s `3d` directory if you want module 3D models in the viewer.

A.2 Importing LCSC/JLCPCB parts (easyeda2kicad)

Several footprints in this project (the `easyeda2kicad:` prefix — the FPC connector, inductor, MOSFET, switches) were generated directly from LCSC’s own part models rather than drawn

by hand:

1. `pip install easyeda2kicad`
2. `easyeda2kicad --full --lcsc_id=C262643` — symbol, footprint and 3D model in one command, per part.
3. Register the generated `easyeda2kicad` library once, the same way as above.

A.3 Fabrication exports

Two working options for JLCPCB production files:

- **Fabrication Toolkit** (install via *Tools* → *Plugin and Content Manager*): one click produces gerbers, BOM and CPL in JLCPCB format, reading each symbol’s LCSC field — see the README section on including component part numbers in production files: <https://github.com/bennymeg/Fabrication-Toolkit>.
- **Native KiCad**, as used for this run: gerbers and drill per the export settings recorded in the change log; CPL via *PCB editor* → *File* → *Fabrication Files* → *Component Placement*; BOM via the *schematic* editor’s symbol-fields export — note the PCB-side BOM export does *not* include custom fields like LCSC. Remember to exclude the DNP rows (C5/C6) and the copper antenna (AE1) before upload.

The `kicad-jlcpcb-tools` plugin (<https://github.com/Bouni/kicad-jlcpcb-tools/releases>) additionally browses the JLCPCB parts database from inside KiCad, and parts can be reserved ahead of an order through the JLCPCB parts cart / private library (<https://jlcpcb.com/user-center/smtPrivateLibrary/partsCart/>).

A.4 Analysis tools used in this design

- **ST eDesign antenna calculator** — coil geometry and inductance iteration: <https://eds.st.com/antenna/>
- **All About Circuits LC resonance calculator** — resonant tank sanity checks: <https://www.allaboutcircuits.com/tools/lc-resonance-calculator/>
- **“Design Your Own NFC Antenna”** — All on electronics (video walkthrough that informed the antenna approach): <https://www.youtube.com/watch?v=j1Ck5NCs40M>
- **Samsung MLCC characteristics viewer** — the DC-bias curves behind the VCOM capacitor verification: <https://weblib.samsungsem.com/mlcc/mlcc-ec.do>

B Specification Sheet

ESP32-C3 e-paper NFC device — rev A — 2026-08-24. Values are marked [V] verified against a datasheet, layout file or CAD model, [E] estimated, or [TBC] to be confirmed on hardware.

B.1 At a Glance

Function	E-paper display with NFC tap-to-share and Wi-Fi/BLE connectivity
Modes	Desk nameplate / event badge / tap-to-share card (firmware-selected)
Outer dimensions	104.6 × 69.6 × 11.1 mm [V]
Mass, assembled	≈ 68 g [E]
Power	1S Li-Po, USB-C rechargeable
Enclosure	Two-part 3D print, M2 screws

Table 7: Device overview.

B.2 Mechanical

B.2.1 PCB

Outline	100.0 × 60.0 mm [V]
Area	6000 mm ² (60 cm ²)
Thickness	1.6 mm [V]
Layers	2 (F.Cu, B.Cu) [V]
Assembly	Single-sided — all components on B.Cu [V]
Mounting holes	4 × Ø2.2 mm NPTH for M2, 2.75 mm in from each corner [V]
Copper features on F.Cu	NFC antenna coil only
Min track width	0.25 mm signal / 0.40 mm power [V]
Min clearance	0.20 mm (0.50 mm on HV class) [V]
Via / drill	0.6 / 0.3 mm [V]
Surface finish	ENIG recommended
Placements	59 across 27 orderable line items [V]

Table 8: PCB mechanical specification.

B.2.2 Enclosure

Outer	104.6 × 69.6 × 11.1 mm [V]
Wall thickness	2.0 mm [V]
Floor thickness	2.0 mm [V]
PCB fit clearance	0.3 mm per side [V], may open to 0.4–0.5 mm [TBC]
Split line	z = +1.6 mm — tray below, cover above [V]
Internal cavity	100.6 × 65.6 × 5.5 mm deep [V]
Board support	4 × Ø4.5 mm bosses with Ø1.7 mm pilot holes [V]
Fasteners	4 × M2 self-tapping; nylon in the two holes nearest the coil
Tray volume	20 591 mm ³ ≈ 25.5 g PLA at 100% infill [V]
Cover volume	9 736 mm ³ ≈ 12.1 g PLA at 100% infill [V]
USB-C opening	12.6 × 6.2 mm [V]
Button access	2 × Ø2.0 mm through the tray floor [V]

Table 9: Enclosure specification.

Case outline is $104.6 = \text{board } 100 + 0.3 \text{ clearance} + 2.0 \text{ wall per side}$. The 69.6 dimension (rather than 64.6) accommodates the ESP32 module’s antenna, which overhangs the board edge by 4.72 mm by design.

B.2.3 Stack-up

Layer	Thickness	z range
Front cover, panel recessed into it	2.0 mm	+1.6 → +3.6
PCB	1.6 mm	0 → +1.6
Battery + clearance	5.5 mm	0 → −5.5
Tray floor	2.0 mm	−5.5 → −7.5
Total	11.1 mm	

Table 10: Vertical stack-up.

B.3 Display

Part	GDEM0266T71WT (Good Display)
Technology	E-paper, monochrome, bistable — zero power to hold an image
Controller	SSD1680 (Solomon Systech)
Diagonal	2.66 in
Resolution	152 × 296 px [V]
Active area	30.704 × 60.088 mm [V]
Outline	36.30 × 71.82 × 1.0 mm [V]
Interface	4-wire SPI, BS1 strapped low [V]
Connector	24-pin 0.5 mm FPC, 14 mm tail [V]
HV rails	Generated on board — VGH, VGL, VSH1/2, VSL, VCOM
Typical operating current	4.5 mA [V]
Peak refresh current	40 mA typ / 60 mA max [V]
Temperature sensing	SSD1680 internal sensor (command 0x18, data 0x80)
Operating temperature	[TBC] — e-paper is expected to be the limiting component

Table 11: Display specification.

The active area is offset 6.07 mm from centre along the long axis, away from the FPC — the enclosure aperture is positioned accordingly, not centred.

B.4 Wireless

MCU module	ESP32-C3-MINI-1-N4 (Espressif), RF pre-certified
Core	32-bit RISC-V single core
Flash	4 MB
Wi-Fi	802.11 b/g/n, 2.4 GHz
Bluetooth	BLE 5
Antenna (Wi-Fi/BLE)	Integrated module PCB antenna, overhanging the board edge per Espressif guidance [V]
USB	Native USB (GPIO18/19) — programming and debug, no UART bridge

Table 12: Wireless MCU specification.

B.4.1 NFC

Tag IC	NT3H2111W0FHKH (NXP NTAG I ² C plus)
Standard	NFC Forum Type 2, 13.56 MHz
Host interface	I ² C
Antenna	Custom PCB coil on F.Cu — 8 turns, 0.6 mm trace, 0.3 mm gap [V]
Coil area	15 × 50 mm [V]
Coil inductance	≈ 1.93 µH [E] — two independent calculation methods agree
Chip capacitance	50 pF [V] (datasheet §13.1)
Tuning capacitor	22 pF C0G/NP0 [V]
Field-detect	FD → GPIO3, deep-sleep wake on tap [V]
Read range	[TBC] — measure after assembly; C14 is the tuning point

Table 13: NFC specification.

Chemistry	Lithium polymer, 1 cell
Capacity	500 mAh (200–600 mAh range supported)
Nominal / max voltage	3.7 V / 4.2 V
Cell	Uxney 503035, 35 × 30 × 5 mm, integrated protection
Connector	JST-PH 2.0 mm, S2B-PH-SM4-TB horizontal SMT [V]
Connector rating	2 A, 100 V, −25 to +85 °C [V]
Retention	Printed rib frame, 0.5 mm clearance all round [V]

Table 14: Battery specification.

B.5 Power

B.5.1 Battery

Polarity is not standardised on JST-PH plugs — verify with a multimeter against J2 before first connection.

B.5.2 Charging — MCP73871-2CC

Parameter	Value	Set by
Input	USB-C VBUS, 5 V	
Regulation voltage	4.20 V [V]	part suffix -2CC
Input current limit	500 mA [V]	PROG2 → OUT
Charge current	≈ 303 mA [E]	PROG1, 3.3 kΩ (Eq 5-
Termination current	≈ 42 mA [E]	PROG3, 24 kΩ (Eq 4-2
Safety timer	6 h [V]	TE → GND
Low-battery threshold	3.1 V [V]	part suffix -2CC
Thermal regulation	THERM held at 0.5 V by fixed 10 kΩ [V]	see note
Worst-case dissipation	≈ 0.4 W [E]	
Topology	Power-path — system runs from USB when present, battery otherwise	

Table 15: Charger configuration.

The battery pack has no accessible thermistor, so THERM is given a fixed 10 kΩ to VSS. The pin sources 50 μA, giving 0.5 V — between the 1.24 V cold and 0.25 V hot fault thresholds. Temperature protection is delegated to the pack’s own circuit.

B.5.3 Regulation — AP2112K-3.3

Output	3.3 V
Rated current	600 mA min [V]
Input source	Charger SYS/OUT rail, not the battery terminal
Package	SOT-23-5
Decoupling	1 μF in, 1 μF out

Table 16: LDO regulator specification.

B.5.4 Fuel gauge — MAX17048

B.5.5 Current Budget [E]

Battery life is not quoted — it depends entirely on refresh interval and radio duty cycle, and should be measured on hardware rather than predicted.

Method	ModelGauge — no sense resistor
Operating current	3 μ A [V]
Interface	I ² C
Sense point	VBAT (VDD and CELL both on the raw cell) [V]
Alert	ALRT $\rightarrow$ GPIO4, deep-sleep wake [V]

Table 17: Fuel gauge specification.

State	Estimate	Note
Deep sleep	$\approx 10\text{--}20\ \mu\text{A}$	MCU deep sleep + gauge 3 μ A + leakage
Display holding image	0 μ A	bistable — no power to retain
Display refresh	40–60 mA for a few seconds	panel datasheet §6.2
Wi-Fi transmit	up to ~ 350 mA peak	ESP32-C3 typical
NFC tap (harvested)	field-powered	tag responds without waking MCU

Table 18: Current budget by state.

B.6 Interfaces and Controls

Item	Part	Notes
USB-C	HRO TYPE-C-31-M-12, 16P	USB 2.0, CC1/CC2 each 5.1 k Ω to GND
Battery	JST-PH S2B-PH-SM4-TB	horizontal SMT, silkscreen “+” marked
Display FPC	Jushuo AFC07-S24ECA-00	24P, 0.5 mm, top contact
Buttons	2 $\times$ TS-1088-AR02016	BOOT and RESET, 2.0 mm actuator, 1.6 N
Status LEDs	3 $\times$ red 0603	STAT1, STAT2, PG — lit only when USB present

Table 19: Interfaces and controls.

B.6.1 Pin Map

Wake sources are restricted to GPIO0–5, the RTC-capable range: GPIO3 for an NFC tap and GPIO4 for a low-battery alert.

B.7 Manufacturing

Pre-order checklist

- Verify polarised-part rotations (Q2, U7, U2, U4, diodes, LEDs) in the assembly preview
- Confirm the BOM excludes DNP rows and the etched antenna
- Dry-fit the FPC before printing the cover — the path has 0.07 mm of margin

B.8 Environmental

B.9 Open Items

1. FPC tail length has 0.07 mm of margin — dry-fit before committing to a print
2. Connector clearances rest on datasheet figures; J1, J2 and J3 have no 3D models
3. The 0.3 mm case fit clearance is untested on a real print
4. NFC read range unmeasured; C14 is the accessible tuning point
5. Board not yet fabricated

GPIO	Net	Note
0	I2C_SDA	shared bus — NFC + fuel gauge
1	I2C_SCL	shared bus
2	EPD_RST	strapping pin, 10 k Ω pull-up
3	NFC_FD	deep-sleep wake (RTC-capable)
4	FUELGAUGE_ALERT	deep-sleep wake (RTC-capable)
5	EPD_BUSY	input
6	EPDCLK	e-paper SPI clock
7	EPDDATA	e-paper SPI data
8	STATUS_LED	strapping, high at boot
9	BOOT	strapping, pull-up + debounce
10	EPD_CS	chip select
18 / 19	USB D- / D+	0 Ω series
20	EPD_DC	data/command
21	UART0_TX	optional debug
EN	RESET	10 k Ω pull-up + cap + button

Table 20: GPIO pin map.

Fabricator	JLCPCB
Process	2-layer, full SMT assembly, single-sided
Design rules	Within standard capability — no fine-feature surcharge
DRC	0 violations, 12 documented exclusions [V]
ERC	Clean apart from one intentional informational note [V]
Gerber set	9 files, Protel extensions, X2 attributes
BOM	27 line items, 59 placements, LCSC part numbers on every row [V]
DNP	C5, C6 — USB ESD positions, deliberately unpopulated
Not in BOM	AE1 — the antenna is etched copper
Enclosure	FDM, PLA, 2 parts, no supports required [TBC]

Table 21: Manufacturing summary.

Operating temperature	[TBC] — limited by the e-paper panel
Connector ratings	JST-PH -25 to $+85$ $^{\circ}\text{C}$, USB-C -30 to $+80$ $^{\circ}\text{C}$ [V]
Battery	charge 0 to $+45$ $^{\circ}\text{C}$ typical for Li-Po [TBC]
Ingress protection	none

Table 22: Environmental ratings.

C Full Bill of Materials

Designators	Value / Part	Package	LCSC
U1	ESP32-C3-MINI-1-N4	module	C2838502
U2	MAX17048G+T10	TDFN-8 + EP	C2682616
U3	AP2112K-3.3	SOT-23-5	C51118
U4	NT3H2111W0FHKH	XQFN-8	C710403
U7	MCP73871-2CC	QFN-20 + EP	C5121473
J1	USB-C TYPE-C-31-M-12	16P	C165948
J2	JST S2B-PH-SM4-TB	PH 2.0 horiz.	C295747
J3	AFC07-S24ECA-00	24P 0.5 mm top-contact	C262643
Q2	SI1304BDL	SOT-323	C7419947
L1	47 μ H CDRH2D18	SMD	C2454210
D1, D5, D6	MBR0530	SOD-123	C77336
D2, D3, D4	Red LED	0603	C2286
SW1, SW2	TS-1088-AR02016	4 $\times$ 3 mm	C720477
R2,R3,R8,R9,R12,R17–R19	10 k Ω	0805	C17414
R1, R4, R5	0 Ω	0603	C21189
R6, R7	5.1 k Ω	0603	C23186
R10	3.3 k Ω	0603	C22978
R11	24 k Ω	0603	C23352
R13, R14, R15	470 Ω	0603	C23179
R16	2.2 Ω	0805	C17521
C1, C3, C8, C24	100 nF	0805	C28233
C2, C11	10 μ F	0603	C19702
C4, C9, C10, C18	1 μ F 50 V X7R	0805	C28323
C7, C12	4.7 μ F	0603	C19666
C13	220 nF	0603	C21120
C14	22 pF C0G	0603	C1653
C15–C17, C19–C23	1 μ F 50 V X5R	0603	C15849
C5, C6	DNP (ESD positions)	0603	—
AE1	NFC coil (etched copper)	—	—

Table 23: Complete BOM, 27 orderable line items / 59 placements.

D References

- Espressif, *ESP32-C3-MINI-1 Datasheet* — module dimensions, peripheral schematics, land pattern.
- Espressif, *ESP32-C3 Hardware Design Guidelines* — module antenna placement.
- Microchip, *MCP73871 Datasheet* + evaluation board layout — strap configuration, THERM behavior (§5.1.4), thermal practice.
- Analog Devices/Maxim, *MAX17048/49 Datasheet* — VDD/CELL sensing.
- NXP, *NT3H2111/2211 Datasheet* §13.1 + *AN11276* — internal capacitance, antenna equivalent circuit.
- Dalian Good Display, *GDEM0266T71WT Product Specification* Rev 1.0 — §6.2 DC characteristics, §9 reference application circuit.
- Solomon Systech, *SSD1680 Datasheet* — §5 pin function, §6.3.1 interface select, command table.
- Sumida, *CDRH2D18/LD Datasheet* — inductor ratings.
- JST, *PH Connector Datasheet* — battery connector ratings.
- Design tools, plugins and calculators: see Appendix A.
- JLCPCB, *Generating Gerber and drill files in KiCad* — export configuration.